

MarketNest

User Stories & Feature Specifications

Complete feature backlog with acceptance criteria — v1.0

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This document defines every feature of the MarketNest platform in the form of user stories. Each story follows the format: As a [role], I want to [action], so that [benefit]. Every story includes acceptance criteria that define the conditions for the story to be considered done. Stories are grouped by system (Super Admin, Seller, Buyer) and feature area.

P1 — Must have (launch)

P2 — Should have (month 2)

P3 — Nice to have (month 4+)

Section 1 — Super Admin System

The Super Admin is the single platform owner. There is exactly one super admin account, bootstrapped via a seed script. The super admin has unrestricted access to all platform data and is the only actor who can create seller accounts.

F-01 Super Admin Authentication	
Secure, session-based login for the super admin. No self-registration — account exists from seed.	
SA-01 As a super admin, I want to log in with my email and password, so that I can access the admin dashboard securely..	Acceptance criteria: ✓ Login form accepts email + password ✓ Invalid credentials show a clear error message without revealing which field is wrong ✓ Successful login creates a JWT session stored in an HTTP-only cookie ✓ Session expires after 8 hours of inactivity ✓ Failed login attempts are rate-limited to 5 per minute per IP ✓ Logout button invalidates the session immediately
SA-02 As a super admin, I want to see a dashboard upon login, so that I get an immediate overview of the platform health..	Acceptance criteria: ✓ Dashboard shows: total sellers, total buyers, total orders today, GMV today ✓ All metrics load within 1 second ✓ Dashboard refreshes data every 60 seconds automatically

F-02 Seller Account Management

Super admin creates, views, suspends, and deletes seller accounts. No seller can self-register.

SA-03 As a **super admin**, I want to create a new seller account by entering their details, so that the seller receives an invitation email to set up their account..

Acceptance criteria:

- ✓ Form requires: store name, seller email, commission rate
- ✓ System sends an invite email with a time-limited link (48 hours)
- ✓ Seller appears in the seller list with status=invited
- ✓ If the invite link expires, admin can resend it
- ✓ Duplicate email address shows a clear validation error

SA-04 As a **super admin**, I want to view a list of all sellers with their key metrics, so that I can monitor seller performance at a glance..

Acceptance criteria:

- ✓ List shows: store name, email, total products, total orders, GMV, status, joined date
- ✓ List is searchable by store name or email
- ✓ List is sortable by each column
- ✓ Pagination: 20 sellers per page

SA-05 As a **super admin**, I want to suspend a seller account, so that their products are hidden from buyers and they cannot log in until reactivated..

Acceptance criteria:

- ✓ Suspend action requires a reason (text field, mandatory)
- ✓ Seller's products immediately become invisible to buyers
- ✓ Seller receives an email notification of suspension with the reason
- ✓ Seller cannot log in while suspended — sees a clear suspended message
- ✓ Admin can reactivate the seller with one click

SA-06 As a **super admin**, I want to permanently delete a seller account, so that their data is removed from the platform..

Acceptance criteria:

- ✓ Delete action requires typing 'DELETE' to confirm
- ✓ Seller's products are soft-deleted (hidden, not removed from DB)
- ✓ Existing orders referencing the seller's products remain intact for audit
- ✓ Action is recorded in the audit log with admin ID and timestamp
- ✓ Deleted seller cannot be recovered — admin is warned of this

F-03 Platform Analytics & Reporting

Real-time and historical view of platform performance.

SA-07 As a **super admin**, I want to view platform-wide revenue and order metrics, so that I can make informed business decisions..

Acceptance criteria:

- ✓ Dashboard shows: GMV (today, 7d, 30d, all-time), order count, average order value
- ✓ Line chart shows daily GMV for the past 30 days
- ✓ Revenue breakdown by category (pie/bar chart)
- ✓ Top 10 sellers by GMV
- ✓ Top 10 products by sales volume

SA-08 As a **super admin**, I want to export analytics data as a CSV, so that I can analyse data in my own tools..

Acceptance criteria:

- ✓ Export button on analytics page
- ✓ Export runs as a background job and emails the CSV download link
- ✓ CSV includes: date, orders, GMV, new users, new sellers, top products
- ✓ Date range is selectable before export

F-04 Category & Catalogue Management

Super admin owns the global category tree that sellers use to classify their products.

SA-09 As a **super admin**, I want to create, edit, and delete product categories, so that sellers have an accurate and organised taxonomy to list their products..

Acceptance criteria:

- ✓ Category has: name, slug (auto-generated), parent category (optional), image, sort order
- ✓ Nested categories up to 3 levels deep
- ✓ Deleting a category with active products is blocked — admin must reassign products first
- ✓ Category changes are reflected on the buyer storefront within 60 seconds (ISR revalidation)

SA-10 As a **super admin**, I want to manage homepage banners and featured listings, so that I can run promotions and highlight specific products..

Acceptance criteria:

- ✓ Banner management: upload image, set link URL, set display order, set active dates
- ✓ Featured products: select up to 8 products to appear in the 'Featured' section
- ✓ Changes go live immediately (no ISR delay for dynamic sections)

Section 2 — Seller System

Sellers are created by the super admin. Each seller manages their own product catalogue, views and processes their orders, and tracks their earnings. Sellers cannot see data from other sellers — this isolation is enforced at the database level with RLS.

F-05 Seller Onboarding	
Seller receives an email invite and completes their account setup.	
SE-01 As a invited seller, I want to click the invite link and set up my account, so that I can start listing products on the platform..	Acceptance criteria: ✓ Invite link opens a setup page with the seller's email pre-filled ✓ Seller sets: password, full name, phone number, store description, store logo ✓ Store slug is auto-generated from store name (editable) ✓ On completion, seller is redirected to their dashboard ✓ Invite link expires after 48 hours — seller sees a clear expiry message
SE-02 As a seller, I want to log in to my seller portal, so that I can access my dashboard and manage my store..	Acceptance criteria: ✓ Email + password login ✓ Forgot password flow sends reset email within 30 seconds ✓ Session lasts 8 hours, with 'remember me' extending to 30 days ✓ Suspended account shows a clear message, not a generic error

F-06 Product Management

Sellers manage their full product catalogue: add, edit, delete, and manage stock.

SE-03 As a **seller**, I want to add a new product to my catalogue, so that buyers can discover and purchase it..

Acceptance criteria:

- ✓ Product form: title, description (rich text), category, price, compare price (optional), SKU, stock quantity
- ✓ Up to 10 images per product, drag-and-drop upload, max 5MB each
- ✓ Images are uploaded to Supabase Storage and a CDN URL is stored
- ✓ Product can be saved as draft or published immediately
- ✓ Published product appears on the buyer storefront within 60 seconds
- ✓ Embedding is generated automatically in the background after publish

SE-04 As a **seller**, I want to add variants to a product, so that buyers can choose the correct size, colour, or configuration..

Acceptance criteria:

- ✓ Variants defined by option names (e.g. Size, Colour) and values (e.g. S, M, L)
- ✓ Each variant can have its own price delta and stock quantity
- ✓ At least one variant can be marked as the default
- ✓ Out-of-stock variants are shown to buyers but not purchasable

SE-05 As a **seller**, I want to edit an existing product, so that I can keep product information accurate and up to date..

Acceptance criteria:

- ✓ All product fields are editable after publish
- ✓ Image reordering via drag-and-drop
- ✓ Changes to price or stock take effect immediately
- ✓ Editing a published product does not take it offline during save

SE-06 As a **seller**, I want to view and manage my product list, so that I have a clear overview of my entire catalogue..

Acceptance criteria:

- ✓ List view with search, filter by category, filter by status
- ✓ Shows: title, thumbnail, price, stock, status, sales count
- ✓ Bulk actions: publish, unpublish, delete selected
- ✓ Low-stock alert: products with stock < 5 are highlighted in amber

F-07 Seller Order Management

Sellers view and process orders that contain their products.

SE-07 As a **seller**, I want to see a list of all orders containing my products, so that I can process them efficiently..

Acceptance criteria:

- ✓ Order list shows: order ID, buyer name, items (from my store), total (my portion), status, date
- ✓ Filterable by status: new, processing, shipped, delivered, cancelled
- ✓ New orders trigger a browser notification and an email alert
- ✓ Orders from other sellers are never visible

SE-08 As a **seller**, I want to update the status of an order, so that the buyer is kept informed of their order progress..

Acceptance criteria:

- ✓ Status transitions: confirmed → processing → shipped → delivered
- ✓ Marking as 'shipped' requires entering a tracking number and courier name
- ✓ Each status change sends an automated email and SMS to the buyer
- ✓ Real-time status update visible to buyer within 2 seconds via Supabase Realtime
- ✓ Cancellation is allowed only in 'confirmed' status — seller must provide a reason

SE-09 As a **seller**, I want to view full order details, so that I have all the information I need to fulfil the order..

Acceptance criteria:

- ✓ Details show: buyer name, shipping address, phone number, items ordered with variants, quantities, prices
- ✓ Payment method shown (COD or online)
- ✓ Order timeline shows each status change with timestamp

F-08 Seller Earnings & Payouts

Sellers track their revenue and upcoming payouts.

SE-10 As a seller, I want to view my earnings dashboard, so that I can track my revenue and understand my commission deductions..

Acceptance criteria:

- ✓ Shows: gross sales, platform commission, net earnings for this week / this month / all-time
- ✓ Bar chart of daily earnings for the past 30 days
- ✓ Commission rate is shown clearly (e.g. 10%)

SE-11 As a seller, I want to view my payout history, so that I have a clear record of all payments received..

Acceptance criteria:

- ✓ Table of all payouts: period, gross, commission, net, status (pending/paid), paid date
- ✓ Downloadable as CSV or PDF
- ✓ Pending payout shows estimated payment date

Section 3 — Buyer / Storefront System

Buyers interact with the public-facing storefront. They can browse and add to cart without an account. An account (via Google OAuth or email) is required only at checkout. Both Cash on Delivery and online payment are supported.

F-09 Product Discovery & Browse

Fast, intuitive product browsing for all users — authenticated or guest.

BU-01 As a **guest user**, I want to browse products without creating an account, so that I can explore the platform before committing..

Acceptance criteria:

- ✓ Homepage loads in under 1 second globally (ISR-cached)
- ✓ All product pages are accessible without login
- ✓ No sign-in prompts during browsing
- ✓ Category navigation is visible and functional from the homepage

BU-02 As a **buyer**, I want to search for products using keywords, so that I can quickly find what I am looking for..

Acceptance criteria:

- ✓ Search bar is prominent on every page
- ✓ Results appear within 300ms
- ✓ Semantic search returns relevant results even with imprecise keywords
- ✓ Spelling corrections are handled gracefully
- ✓ Search results show: product image, title, price, seller name, rating

BU-03 As a **buyer**, I want to filter and sort search results, so that I can narrow down results to exactly what I need..

Acceptance criteria:

- ✓ Filters: category, price range (min/max slider), rating (min stars), seller
- ✓ Sort options: relevance, price low-to-high, price high-to-low, newest, best rated
- ✓ Filters are applied instantly without a full page reload
- ✓ Active filters shown as removable chips

BU-04 As a **buyer**, I want to view a product detail page, so that I have all the information needed to make a purchase decision..

Acceptance criteria:

- ✓ Shows: all product images (gallery), title, price, compare price (with % discount badge)
- ✓ Variant selector: size, colour etc. with clear stock status
- ✓ Full product description with rich text formatting
- ✓ Seller name with link to seller storefront page
- ✓ Average rating and review count
- ✓ Related products section (AI-powered)

F-10 Shopping Cart

Guest and authenticated cart with seamless merge on login.

BU-05 As a **guest user**, I want to add products to my cart without logging in, so that I can shop and decide to check out later..

Acceptance criteria:

- ✓ Add to cart works immediately — no login prompt
- ✓ Cart icon in header shows item count in real time
- ✓ Cart persists for 7 days in Redis using my session token
- ✓ Adding the same product again increases the quantity
- ✓ If product goes out of stock after adding, cart shows a warning at checkout

BU-06 As a **buyer**, I want to view and manage my cart, so that I can review my selections before purchasing..

Acceptance criteria:

- ✓ Cart page shows: product image, title, variant, unit price, quantity selector, line total, remove button
- ✓ Quantity can be changed inline — total updates immediately
- ✓ Stock validation: if requested quantity exceeds stock, quantity is capped with a message
- ✓ Order summary shows: subtotal, estimated shipping, total
- ✓ Empty cart shows a friendly prompt to start shopping

BU-07 As a **buyer**, I want to have my guest cart merged with my account after I log in, so that I do not lose the items I added before signing in..

Acceptance criteria:

- ✓ On login at checkout, guest cart items are added to the user's existing cart
- ✓ If the same product exists in both carts, quantities are summed (capped at stock)
- ✓ Guest session token is invalidated after merge
- ✓ Merge is invisible to the user — they see a unified cart

F-11 Checkout & Payment

Streamlined checkout supporting both COD and online payment.

BU-08 As a **guest user**, I want to be prompted to log in or create an account only at checkout, so that I can browse freely but still complete a purchase securely..

Acceptance criteria:

- ✓ Checkout page is the first and only point where login is required
- ✓ Clear messaging: 'Sign in to complete your order'
- ✓ Google OAuth one-click sign-in available
- ✓ After sign-in, buyer is returned directly to checkout — not homepage
- ✓ Guest cart is merged automatically on login

BU-09 As a **buyer**, I want to enter my shipping address at checkout, so that my order is delivered to the correct location..

Acceptance criteria:

- ✓ Address form: full name, phone, address line 1, address line 2, city, state, postal code, country
- ✓ Saved addresses for returning buyers — select from list or add new
- ✓ Phone number is mandatory (needed for delivery coordination)
- ✓ Address is validated for completeness before order is placed

BU-10 As a **buyer**, I want to choose between Cash on Delivery and online payment, so that I can pay in the way that is most convenient for me..

Acceptance criteria:

- ✓ Payment method selector clearly shows both options
- ✓ COD option shows estimated delivery time
- ✓ Online payment opens Stripe Elements (card number, expiry, CVC)
- ✓ Apple Pay and Google Pay shown if supported by the device
- ✓ Selected payment method is displayed on the order summary

BU-11 As a **buyer**, I want to complete an online payment securely, so that my order is confirmed and I receive a receipt..

Acceptance criteria:

- ✓ Stripe Elements embedded in page — no redirect for card payment
- ✓ Payment processing shows a loading state — button disabled to prevent double-submit
- ✓ Successful payment immediately redirects to order confirmation page
- ✓ Order confirmation email sent within 30 seconds
- ✓ Failed payment shows a specific error message (e.g. 'Card declined') with retry option

BU-12 As a buyer , I want to place a Cash on Delivery order, so that my order is registered and the seller is notified..	<i>Acceptance criteria:</i> ✓ COD order is created immediately on form submission ✓ Order confirmation page confirms: order ID, items, total, delivery estimate ✓ Seller receives email notification of the new COD order ✓ Buyer receives order confirmation email ✓ Order status starts as pending_cod
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F-12 Order History & Tracking

Buyers can track all past and current orders in real time.

BU-13 As a buyer , I want to view my order history, so that I can track all my past and current orders..	<i>Acceptance criteria:</i> ✓ Order list shows: order ID, date, items summary, total, status badge ✓ Filterable by status ✓ Clicking an order opens the detail view
BU-14 As a buyer , I want to track my order status in real time, so that I always know where my order is..	<i>Acceptance criteria:</i> ✓ Order detail shows a visual timeline: confirmed → processing → shipped → delivered ✓ Tracking number and courier name shown once order is shipped ✓ Status updates push via Supabase Realtime — page updates without refresh ✓ Email and SMS notification at each status change
BU-15 As a buyer , I want to cancel an order that has not yet been processed, so that I am not charged for an order I no longer want..	<i>Acceptance criteria:</i> ✓ Cancel button visible while order is in 'confirmed' status ✓ Cancellation confirmation prompt before final action ✓ Online payment: refund initiated via Stripe within 5 minutes ✓ Refund timeline communicated clearly (3-5 business days typical) ✓ COD order: cancelled without any charge

F-13 Buyer Account & Profile

Buyers manage their personal information, addresses, and account security.

BU-16 As a **buyer**, I want to create an account using Google, so that I can sign up with one click without remembering a new password..

Acceptance criteria:

- ✓ 'Continue with Google' button on sign-in and checkout pages
- ✓ First Google sign-in creates a profile with role=buyer automatically
- ✓ Subsequent sign-ins with the same Google account always route to the same profile
- ✓ If an email account already exists with the same email, accounts are merged

BU-17 As a **buyer**, I want to update my profile information, so that my account details stay current..

Acceptance criteria:

- ✓ Editable fields: full name, phone number, avatar (upload)
- ✓ Email is non-editable (tied to auth provider)
- ✓ Password change available for email/password accounts
- ✓ Changes saved with confirmation toast

BU-18 As a **buyer**, I want to manage multiple saved addresses, so that I do not have to re-enter my address for every order..

Acceptance criteria:

- ✓ Add, edit, delete saved addresses
- ✓ Set a default address
- ✓ Maximum 5 saved addresses per account
- ✓ Default address pre-selected at checkout

F-14 Product Reviews & Ratings

Buyers leave reviews on products they have purchased.

BU-19 As a **buyer**, I want to leave a rating and review for a product I purchased, so that other shoppers benefit from my experience..

Acceptance criteria:

- ✓ Review prompt appears after order is marked as delivered
- ✓ Star rating 1-5 is mandatory
- ✓ Written review is optional (max 500 characters)
- ✓ Only verified purchasers can leave reviews
- ✓ One review per buyer per product
- ✓ Review appears on product page within 60 seconds

BU-20 As a **buyer**, I want to read reviews from other buyers, so that I can make an informed purchase decision..

Acceptance criteria:

- ✓ Reviews shown on product detail page below the description
- ✓ Average star rating shown prominently near the price
- ✓ Reviews sorted by most recent by default
- ✓ Rating distribution bar chart (5★ 4★ 3★ 2★ 1★ breakdown)

Section 4 — Cross-Cutting & Platform Features

F-15 Notification System

Multi-channel notifications across email, SMS, and real-time in-app for all actors.

PL-01 As a **buyer**, I want to receive an email confirmation for every order I place, so that I have a record of my purchase..

Acceptance criteria:

- ✓ Email sent within 30 seconds of order creation
- ✓ Email includes: order ID, itemised list, totals, estimated delivery, tracking link (once shipped)
- ✓ HTML email with platform branding
- ✓ Plain text fallback included

PL-02 As a **seller**, I want to be notified immediately when a new order comes in, so that I can start processing it without delay..

Acceptance criteria:

- ✓ Email notification within 30 seconds of new order
- ✓ Browser push notification if seller is in the seller portal
- ✓ Notification includes: order ID, items, buyer shipping city, payment method

PL-03 As a **buyer**, I want to receive an SMS when my order ships, so that I know my order is on its way even if I am not checking email..

Acceptance criteria:

- ✓ SMS sent when seller marks order as 'shipped'
- ✓ SMS includes: tracking number and a short tracking URL
- ✓ SMS uses the phone number on the order's shipping address

F-16 AI-Powered Search & Recommendations

Semantic product search and personalised recommendations powered by pgvector and RAG.

PL-04 As a **buyer**, I want to find products using natural language, so that I get accurate results even when I do not know the exact product name..

Acceptance criteria:

- ✓ Search uses vector similarity on product embeddings
- ✓ Results are relevant even with synonyms or descriptive queries
- ✓ Keyword search remains as a fallback if vector search returns < 3 results
- ✓ Search latency < 300ms at the 95th percentile

PL-05 As a **buyer**, I want to see personalised product recommendations, so that I discover products I am likely to enjoy without having to search..

Acceptance criteria:

- ✓ 'Recommended for you' section on homepage (logged-in users)
- ✓ Recommendations based on purchase history and browsing events
- ✓ Updated daily via background job
- ✓ New users see 'Trending' products until personalisation is available

PL-06 As a **buyer**, I want to use an AI chat assistant to find products, so that I can describe what I want in plain language and get curated suggestions..

Acceptance criteria:

- ✓ Chat widget available on the buyer storefront
- ✓ Natural language product discovery ('gift for my mum who likes cooking under \$50')
- ✓ AI retrieves top matching products via RAG and presents them as clickable cards
- ✓ Conversation history maintained within the session

F-17 Performance & Reliability

Non-functional requirements captured as user-facing acceptance criteria.

PL-07 As a **buyer**, I want to have the storefront load quickly on any device, so that shopping is a pleasant experience regardless of my connection speed..

Acceptance criteria:

- ✓ Homepage Largest Contentful Paint (LCP) < 1.5s on 4G
- ✓ Lighthouse Performance score > 90 on desktop and mobile
- ✓ Product images served in WebP format with correct sizing for the device
- ✓ First meaningful content visible within 0.8s

PL-08 As a **buyer**, I want to have the platform remain available during high traffic periods, so that I can shop during sales events without the site going down..

Acceptance criteria:

- ✓ System handles 1,000 concurrent checkouts without degradation
- ✓ API p95 response time < 200ms under normal load
- ✓ API p95 response time < 500ms under 10x peak load
- ✓ CDN absorbs product page traffic — origin sees < 10% of total requests